

BAIE-COMEAU, QC, Canada

WMO: 716910

Lat: 49.1351N

Lon: 68.2024W

Elev: 71

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.6	-10.8	-28.6	1.1	-14.6	-23.4	1.5	-9.9	29.4	26.0	26.4	23.1	8.8	270	0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	75.4	63.4	72.7	62.2	70.1	60.9	66.0	72.3	64.1	70.1	62.4	68.0	13.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.5	87.8	69.0	61.6	82.2	66.8	60.0	77.5	64.9	30.8	72.3	29.4	70.2	28.1	68.1	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.6	22.6	20.0	-23.4	82.6	4.7	2.9	-26.7	84.7	-29.5	86.4	-32.1	88.1	-35.5	90.2	
			-23.8	70.1	4.6	2.2	-27.1	71.7	-29.8	72.9	-32.4	74.2	-35.7	75.7	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	9.4	11.4	21.2	33.7	45.4	55.3	61.1	59.8	51.7	41.4	30.6	18.0
	DBStd	19.49	12.05	11.48	10.20	6.33	5.52	5.42	4.00	4.33	5.57	5.77	8.02	10.44
	HDD50	5796	1260	1082	891	489	167	12	0	1	45	274	583	992
	HDD65	10349	1725	1502	1356	939	609	294	133	168	400	733	1033	1457
	CDD50	944	0	0	0	0	23	171	343	305	96	6	0	0
	CDD65	22	0	0	0	0	0	3	12	7	0	0	0	0
	CDH74	150	0	0	0	0	3	32	69	41	5	0	0	0
	CDH80	12	0	0	0	0	0	4	6	2	0	0	0	0
Wind	WSAvg	9.3	10.0	10.0	10.1	9.7	8.9	8.8	8.4	8.1	8.5	9.3	9.6	9.8
Precipitation	PrecAvg	41.1	3.5	2.8	3.0	3.2	3.5	3.3	3.1	3.6	4.0	3.5	3.9	4.0
	PrecMax	51.2	7.8	5.5	6.3	8.2	7.4	7.3	5.7	7.7	6.5	5.3	7.0	8.5
	PrecMin	32.1	0.9	0.4	0.8	0.4	0.4	1.2	1.3	1.5	1.8	1.0	2.0	1.8
	PrecStd	5.9	1.6	1.3	1.5	1.9	1.7	1.4	1.3	1.5	1.4	1.2	1.4	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	38.1	42.9	53.4	70.8	78.0	79.6	78.6	72.9	61.4	50.5	40.4
		MCWB	35.0	35.5	36.1	43.7	57.1	62.2	66.3	65.1	62.7	53.2	47.6	38.8
	2%	DB	34.2	35.2	38.7	48.1	64.9	73.0	75.6	74.2	67.5	56.5	46.0	37.2
		MCWB	32.9	33.0	34.2	41.0	52.9	60.1	64.2	63.6	59.4	51.1	43.0	35.2
	5%	DB	31.2	31.9	36.4	44.6	59.3	69.5	73.1	71.5	64.0	52.9	43.1	35.0
		MCWB	29.7	30.1	33.1	38.5	49.9	58.5	63.0	62.5	56.9	48.4	40.9	33.5
	10%	DB	27.5	28.9	34.5	42.0	55.2	66.2	70.4	68.7	61.0	50.1	41.0	32.9
		MCWB	26.1	27.1	31.8	37.1	47.9	56.8	61.6	60.7	55.4	46.3	39.0	31.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	36.4	37.9	45.5	59.1	65.9	69.6	67.5	64.7	55.6	48.6	39.2
		MCDB	37.0	37.6	40.7	51.4	66.8	72.9	75.7	74.2	69.9	58.7	49.7	39.5
	2%	WB	33.0	33.3	35.3	42.1	54.0	62.8	66.6	65.4	60.8	52.1	43.7	35.7
		MCDB	34.1	34.5	37.4	46.5	61.9	69.7	72.8	71.6	65.9	55.5	45.0	36.6
	5%	WB	29.7	30.3	33.7	39.6	51.4	60.4	64.7	63.6	58.3	49.5	41.2	33.7
		MCDB	30.9	31.9	35.8	43.5	57.3	67.0	70.8	69.3	62.4	52.2	42.6	34.7
	10%	WB	26.1	27.3	32.0	37.8	48.9	58.1	62.8	62.1	56.4	47.1	39.1	31.8
		MCDB	27.5	28.8	34.1	40.9	54.0	63.9	68.6	67.2	60.1	49.3	40.5	32.8
Mean Daily Temperature Range	5% DB	MDBR	16.5	17.1	15.3	12.4	15.5	17.0	16.7	16.7	15.9	13.0	11.4	13.9
		MCDBR	16.2	12.6	13.3	16.3	24.6	24.3	21.8	21.3	20.1	17.2	11.8	11.2
		MCWBR	16.1	11.9	10.9	11.5	15.1	13.4	12.2	12.2	13.2	12.3	10.4	11.1
	5% WB	MCDBR	15.8	11.7	11.8	14.8	20.8	20.5	19.4	18.3	17.7	15.3	10.8	11.2
		MCWBR	15.8	11.3	10.4	11.0	13.9	12.8	12.2	11.4	12.7	11.9	10.3	11.2
Clear-Sky Solar Irradiance	taub		0.267	0.274	0.305	0.360	0.373	0.398	0.425	0.403	0.378	0.333	0.313	0.271
	taud		2.342	2.318	2.292	2.268	2.340	2.347	2.304	2.367	2.411	2.512	2.393	2.381
	Ebn at Noon		253	280	288	281	281	274	264	265	260	256	231	236
	Edh at Noon		21	28	34	39	38	38	39	35	30	23	21	18
All-Sky Solar Radiation	RadAvg		390	698	1091	1430	1649	1746	1674	1432	1087	639	344	260
	RadStd		43	70	82	119	139	131	165	87	77	55	35	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+3.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (16 neighbors)	N/A	N/A	+2.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page